What survives on a small team's rig: re-testing 21 robot-learning papers with cheap hardware and little data

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Abstract. Most of the robotic papers are either tested in sim or on expensive rigs. From our tested papers, 16 out of 21 were done in simulation or on rigs above $20k, and only 1 on a rig under $5k. But the question is, does a claim still validate for smaller teams with cheaper hardware, less data and shorter tasks? This is what we want to answer in this paper. For that we re-tested 36 claims out of 21 papers on our cheap hardware, an Nvidia Jetson Orin NX 16GB and a robot arm SO-101 (about €1,500). We used the same protocol for each claim: a bar fixed before any testing, and the cheapest baseline tested first. Out of the 36 claims, only 11 had a cheap baseline to test against, and among them only 4 are worse than this baseline. Among those 4, LeWorldModel, DINO-WM and ContactGuard are clean fails, and Robometer's loss comes from how its dataset ends the episodes. LeWorldModel and DINO-WM picked the chunk the arm really executed less often than a baseline using the actions alone, at every distance we tested, and ContactGuard predicted failures worse from its imagined latent than from the current one. Our results: 12 claims hold, 10 fail using the paper's method, 5 fail on our variation, 2 on our inference engine and 1 on a task outside the paper's regime. Most of the failures come as much from the papers' methods as from our short tasks, small data and our own variants, and for each we explain why. The code, data and all numbers are public and can be replicated.

Code and rows: `github.com/ravediamond/paper-register-reproducibility` **Reproduce:** `make check`. Every number carries its source file and protocol commit.

1 Introduction

Out of the 21 papers we tested, 16 of them are done in simulation or on expensive rigs above $20k, and only 1 on a rig under $5k. This means small teams can either trust the paper or re-test the claim on their own hardware. Both of them are costly, either when validating or debugging if it does not work. The question really is, do the papers' claims still hold on cheap hardware, with little data and short tasks? And this is the environment small teams or solo founders have.

To answer this question, we re-tested each claim on our own cheap rig. We used the same testing protocol each time: a bar fixed before any test, and the cheapest baseline (low cost or free) run first on the same data, to filter out claims early before investing in them.

We found out that this cheap baseline was really useful to validate claims. For example with LeWorldModel, the goal was to predict the next state of the arm and plan with it. For that we had a first baseline where it needed to beat a copy-last method, and it passed (error 0.31 vs 0.89). But for ranking the chunk the arm really executed, the model scored worse than using the actions alone (0.60 vs 0.68–0.90). Without that second baseline, we would have called the paper a success.

Out of the 36 claims, only 12 hold on our rig. The failures are split between the paper's method failing (10), our variant of the method failing (5), our inference engine (2) and one outside the paper's regime. But, and it is important to say it again, a fail on our rig does not mean the paper is wrong: it just does not work on our specific setup and data.

2 Rig and data

Here's the rig we used for re-tests:

- A robot arm SO-101: 5 joints + gripper, 1 wrist camera and 1 front-top camera
- Leader + follower arm: all teleoperation done using the leader arm
- A Nvidia Jetson Orin NX 16GB: runs the inference
- Cloud GPU instances for training only

The estimated cost of the whole rig is about €1,500, plus under $50 of cloud GPU (for all re-tests).

Here's the data we used:

The teleoperated data (HF dataset: so101_bowls_all_trimmed):

- 2 short tasks: put one bowl into another
- 182 episodes, 61,462 frames at 30 fps, about 34 min
- split: 157 episodes in the train set, 25 in the validation set
- mostly used for the world models and the VLA policies

Some policy rollouts:

- 69 episodes: 44 successes, 25 failures, labelled by hand by the operator
- main set for most monitor tests: 50 episodes (42 successes / 8 failures), leaving out the episodes of one policy family (TurboVLA's)
- FAIL-Detect / SAFE: the 44 successes split in half, 22 to calibrate and 22 to test, over 20 random splits
- a separate set of 88 of our own episodes, top camera only, to check the reward model directly on our rig

The ArmnetBench dataset (public, HF dataset: armnet/armnetbench_v01_lerobot_so101) [14]: 2,099 labelled SO-101 rollouts from 7 policies, used for the reward models and judges.

The timing set:

- 20 recorded episodes
- 254 chunk boundaries

This is used for analysing and measuring chunking and latency.

But this data has limits that explain some of the failures:

- Short tasks only: each episode is about 11 s
- No multi-step, memory-based or long-horizon tasks
- Everything has been done in one place, with one robot and one task family, which means this may not translate to real-world usage of such tasks.
- All the rollout outcomes have been judged by one human, so they are subjective (not blind).

3 What we tested and how

There are 4 kinds of papers we picked for re-tests (see Table 1 for details):

- World models (6 papers): the goal is to learn a latent from the cameras to predict or plan. In this case, the question is "does the latent beat a free signal?"
- Failure monitors and judges (8 papers): the goal is to detect if an episode fails, or how far it has progressed. In this case, the question is "does the monitor beat a trivial signal?"
- VLA policies (3 papers): the goal is to follow and execute an instruction on the arm.
- Chunking and edge inference (4 papers): the goal is to improve how action chunks are made and joined, by making it smoother or faster.

All the papers were chosen because we judged that they could reasonably run on our rig, which filters out many papers from the get-go.

We used 2 rules to re-test the claims.

Rule 1: the bar to achieve to validate the claim needs to be written from the paper and its code, not after running the claim once. This avoids contaminating the bar with our own results. Where the paper gives a number, the bar comes from it; otherwise the bar is ours.

Rule 2: we create and launch the cheapest baseline first, on the same data. It is built from cheap things that can be used on our tasks, like the last frame, the joint values or the episode length. If the method loses to it, the bar is not useful. For example, does a failure monitor catch failures better than directly checking the episode length?

There are only 4 verdicts possible for each re-test: holds, meaning the bar is met; fails, so the bar is missed; not proven, because we cannot decide considering our rig and context; and not tested, not run yet (because of time to test, cost, ...). A paper is considered failed if any of its claims fails. We decided that because, for us, a paper that is only half successful is considered failed. All the failures can be categorized in 4 categories:

- the paper's method failed
- our variant of the method failed (because we could not use the paper's method directly)
- the limitation of our inference engine
- or a task that goes outside the paper's scope

We had 2 cases of weaker bar: for SmolVLA and VLA-JEPA, we had no numeric bar, so instead we launched 10 trials judged by a human. 9 out of 10 were a success, which means a true rate anywhere from 60% to 98% (Wilson interval, 95%).

4 Results

Table 2 and Figure 1 show the verdicts, and Table 1 breaks them down by kind of paper. Out of the 36 tested claims, 12 hold, 18 fail (10 using the papers' methods, 5 using our variant, 2 on our inference engine and 1 outside the paper's scope), 3 are not proven and 3 not tested. At the paper level, 4 are validated, 14 fail and 3 are not proven.

The pattern by kind looks revealing (all 8 failure monitors and judges have failed, all 3 VLA policies reproduce) but is actually limited. 4 out of 8 monitors used our own variant, so they cannot directly serve as evidence. As for the VLAs, for 2 of the 3 there were no numbers to use as bars, so we just ran trials and recorded the number of fails.

5 The cheapest baseline

The cheapest baseline is the signal we get for free on the same data as the method (Table 1). It needs no training and runs in minutes. We compared 11 paper claims with such a baseline (Table 3). The method clearly beat it in 4 cases, beat it by less than its bar in 2, and lost to it in 5. We ran a paired bootstrap over episodes (2,000 resamples, 95 % percentile interval, protocol frozen before any number) on the 11 of the 11 rows whose per-episode rows were stored or could be written again by re-running the evaluation; for 3 of them the re-run gave back the stored number only to the second or third decimal, and the interval is that of the re-run. For Robo-JEPA the encoder of the original run is not on our machine. 4 losses have their interval below zero: LeWorld-Model and DINO-WM against the actions alone, at every target distance we tested, ContactGuard against the current latent, and Robometer, which comes from how the dataset ends its episodes (below). The intervals of adaptive chunking and of ReAP against shuffled actions contain zero, and LeWorldModel against copy-last, HWM, PACT-VLA and REMORA beat their baselines. LeWorld-Model, DINO-WM and ContactGuard are therefore the three clean cases of a method losing to a free baseline with the authors' method on our data; ContactGuard's paper reports the opposite (0.98 against 0.84).

The choice of baseline decides the verdict, even for a single model. LeWorldModel predicts the next latent far better than copy-last, with an error of 0.31 against 0.89, and that claim holds. Yet the same model ranks the chunk that was really executed worse than the actions alone do, 0.60 against 0.68–0.90, and that claim fails. Predicting well is not the same as planning well, and testing

Table 1: The four kinds of papers, what we test in each, the cheapest baseline we build for it, and the verdicts.

kind of paper	papers	what we test	cheapest baseline	verdicts
world models	6	predict the next state; rank or plan actions	copy-last (next state = current state); the commanded joints or the actions alone	1 reproduce, 4 fail, 1 not proven
failure monitors and judges	8	catch failures; track progress	episode length; the same model with shuffled actions	8 fail
VLA policies	3	succeed on our arm	none: success over 10 trials judged by the operator, no numeric bar fixed before	3 reproduce
chunking and edge inference	4	smoother joins between chunks; time per chunk	the same policy without the trick; the real-time budget on the Orin	2 fail, 2 not proven



Figure 1: the 36 paper claims by verdict: 12 reproduce, 10 fail with the authors' method on our data, 8 fail only in our variant of the method, because of our inference engine or outside the paper's regime, and the rest are open.

Table 2: Verdicts by paper and by claim. A paper fails if any of its claims fails; failures are split by whose method failed.

verdict	papers	claims
reproduces	4	12
fails, authors' method	8	10
fails, our variant	4	5
fails, our engine	1	2
fails, outside the paper's regime	1	1
not proven	3	3
not tested	0	3

only one of the two would give the wrong picture.

The same baseline is also not equally strong on all data. On ArmnetBench, episode length beats the reward model (0.938 against 0.818), because there a failed episode tends to run until the time limit, so the length gives the label away. On our own 88 episodes, where failures do not run to the time limit, length is close to chance (0.521). What a judge must beat therefore depends on who stopped the episode, and the baseline has to be computed on one's own data.

Copy-last and the joints are strong partly because our task is short and simple: little moves apart from the arm, and the arm's state is already in the joints. On long tasks with many moving objects a learned latent has more to add, and we did not test that. Some papers report a cheap baseline themselves (shuffled actions for ReAP, a random pick for PACT-VLA, net joint displacement for HWM); on our data PACT-VLA and HWM beat it, and ReAP's margin over shuffled actions is within the interval; Robometer reports no episode-length baseline. In practice, the baseline is the first thing to run: if the method loses to it, one can stop there, and only if it wins is the bar worth checking.

6 A testing order

The failures point to three checks, in this order. First, time the method on the target device: adaptive chunking needs 4,751 ms for one decision against a budget of 298 ms, partly because our engine samples without batching, and this is the cheapest failure to find. Second, compare it with the cheapest baseline on one's own data: this is where the world models fail, since the commanded joints already hold the arm's state on a short task, and it is also where a dataset can mislead, as episode length does on ArmnetBench. Only then is it worth checking the method against its bar. Of the three, only the learned-latent pattern rests on more than one case, so we offer this as a working order, not as a taxonomy of how methods fail.

7 Why they failed here: the method, our data, or our setup

For each failure, Table 4 says whose method failed and what on our side may explain it. Most failures have at least one such cause, so a failure here is a failure on this rig, with this data, and not a verdict on the method in general.

Our data is small and simple. We have 157 training episodes of one short task, about 11 s each, where the papers use hours of varied data; HWM, for instance, uses about 130 h against our 0.6 h. On a short task with one arm, the commanded joints already hold most of the state, which leaves little for a latent to add. The task is also easy for our policy, since SmolVLA is at its ceiling on it, so there is little uncertainty for an entropy rule to read: the chunk length chosen by adaptive chunking sat at its minimum in 86 % of decisions. For H-JEPA, our

Table 3: The paper claims we compared with a cheap baseline, on the same data.

method	baseline (what is measured)	method	baseline	outcome	paired difference, positive = method better [95 % interval]
LeWorldModel	copy-last (latent error at 30 steps, lower is better)	0.31	0.89	beat	+0.58 [+0.51, +0.66], beats, on the re-run (unit = window)
LeWorldModel	actions alone (executed chunk ranked first)	0.60	0.68–0.90	lost	worst case d = 10: −0.37 [−0.42, −0.32]; best case d = 5: −0.23 [−0.29, −0.18], loses
DINO-WM	actions alone (executed chunk ranked first)	0.61	0.68–0.90	lost	worst case d = 15: −0.28 [−0.31, −0.25]; best case d = 5: −0.10 [−0.15, −0.04], loses, on the re-run
RoboJEPA	copy-last (9-step latent error)	11.8 % below copy-last	–	beat, under its bar of 15 %	+0.12 [+0.11, +0.13], beats
ContactGuard	the current latent (failure AUROC)	0.70	0.82	lost	−0.12 [−0.24, −0.01], loses
ReAP	shuffled actions (failure AUROC)	0.789	0.708	beat	+0.08 [−0.05, +0.20], unclear, on the re-run
Robometer	episode length (failure AUROC, ArmnetBench)	0.818	0.938	lost	−0.12 [−0.15, −0.09], loses
HWM	net joint displacement (cosine to the macro-action)	0.608	0.551	beat	+0.06 [+0.03, +0.08], beats
Adaptive chunking	fixed chunk length (boundary jump, lower is better)	0.973	0.940	lost	−0.03 [−0.17, +0.06], unclear
PACT-VLA	a random pick (error of the executed window)	−0.27° against random	–	beat	+0.27 [+0.09, +0.44], beats
REMORA	per-frame readout (readout flips)	0.70 × per-frame	–	beat, under its bar of 0.5 ×	ratio 0.70 [0.48, 0.94], beats

task lies outside the regime the paper targets, as there is no gap between slow and fast parts of the scene, so we count its failure as out of scope rather than a failure of the method.

We also have too few failures. Our policies mostly succeed, and the main rollout set has only 8 failures in 50 episodes. A bar such as five failures caught out of eight is coarse, and a single episode can change the verdict (ReAP, VLA-Corrector, SURVEYOR).

In 4 papers our version of the method is not exactly theirs. (ContactGuard and the threshold claim of SURVEYOR are counted as the authors' method: we kept their recipe and only fitted it to our rig, with two cameras instead of three and mixed policies for ContactGuard. VLA-Corrector is counted the same way: the features, network, label window and gate are the paper's, and we changed the data, which is 30 training rollouts per fold instead of 120 episodes, with the failure onsets read from video and no stage head.) For FAIL-Detect and SAFE we used our world model's surprise as the score instead of their own scores; their code needs the internal features of the policy that ran each episode, and only 24 successes and 6 failures of ours could be re-run through it. Our first conformal band was also built wrong, calibrated on the same episodes it was fitted on; with the band built as the papers build it, 9.1 % of successes are flagged at α = 5 % (not 22.7 %) and 24 % of failures are caught, and both claims still fail. ReAP's risk head ran on our world model's latent. For SURVEYOR [25], the drafter is not released, so its monitor claim ran with our own model's imagination, and REMORA's progress atoms are written by hand. These are failures of our variants, and we do not claim that the authors' methods fail.

Our inference engine matters as well. vla.cpp samples sequentially and has no batching, so adaptive chunking pays for 20 full passes, while the paper's 106 ms was measured with batching on an A800. For PACT-VLA, the time of the check is projected rather than measured. Finally, several tests were offline only: planning ranked recorded chunks, and chunking measured smoothness on recorded episodes. None of these ran in closed loop on the arm.

8 Limitations

Our data covers one short task family, about 11 s per episode, with one arm, two cameras and about 34 minutes of demonstrations, so it says nothing about long-horizon tasks or varied scenes. There are few failures to test the monitors on, 8 in the main set of 50. Many methods are our own variants rather than the authors' code end to end (Table 4), and the planning and chunking tests ran offline, never in closed loop on the arm. A single operator ran the arm and labelled the outcomes without being blind to them, many tests use small samples and a single training seed, and few results come with a confidence interval; a bootstrap over episodes for the baseline comparisons could be run for only 11 of 11, and 3 of these intervals come from a re-run that matches the stored result only to the second or third decimal. Two VLA results have no numeric bar and rest on 10 trials. The bars are ours, not the papers'. We chose the papers

Table 4: Each failed or not-proven paper, whose method failed, what we measured, and what on our side may explain it.

failed paper	whose method	our result	what may be on our side
LeWorldModel	authors'	planning rank 0.60, action-only 0.68–0.90	one short task, 157 episodes, one seed; offline ranking only
DINO-WM	authors'	planning rank 0.61, action-only 0.68–0.90	same data as LeWorldModel; one training
RoboJEPA	authors'	11.8 % below copy-last, bar 15 %	near miss; the 1B model reached 15.8 % on 30 clips only
H-JEPA	outside its regime	progress ρ −0.73, bar −0.80; false stalls 52 %	our task is outside the paper's regime
HWM	not proven	not proven: high level worse at 2 s	0.6 h of one task against their 130 h
ContactGuard	authors'	AUROC 0.70, current latent 0.82	mixed policies and 2 cameras against one policy and 3 cameras
FAIL-Detect and SAFE	our variant	9.1 % of successes flagged at α 5 % (papers' band)	our score, not theirs; 11 + 11 successes to build the band
ReAP	our variant	3 of 8 failures caught	their risk head on our world model's latent, not their features; 8 failures
VLA-Corrector	authors'	1 of 8 caught at 3 of 42, AUROC 0.53	30 training rollouts per fold against their 120 episodes; our onset labels, no stage head; 8 failures
Robometer	authors'	AUROC 0.818, length 0.938	zero-shot only; ArmnetBench failures run to the time limit (on our rollouts length is near chance)
SURVEYOR	authors' (threshold); our variant (monitor)	threshold 1.249 against 0.20; 3 of 8	their drafter not released; 8 failures
REMORA	our variant	readout flips 0.70 of the per-frame readout	hand-written progress atoms
Adaptive chunking	authors' (latency: our engine)	4,751 ms per decision; length 2 in transport and at the grasp	sequential engine, no batching; task at ceiling for our policy
PACT-VLA	our engine	37 ticks of delay against 8	check time projected; sequential engine
Real-Time Chunking	not proven	not proven: 71–73 % smoother, no arm run	offline smoothness only
Horizon × depth	not proven	not proven: 0.78, bar 0.70	one policy, one task; arm test not run

from the stack we run, so the rates describe these papers and not the field. Finally, our own mistakes are possible.

9 Related work

Two works evaluate policies on the same arm: a real-world VLA benchmark with a taxonomy of failures and recoveries [13], and ArmnetBench, a fleet of SO-101 cells that evaluates policies at scale [14]. In simulation, vla-eval runs VLAs across 14 benchmarks [8], and another benchmark studies the quality of demonstrations [20]. These works rank policies. We instead take the claim a paper makes about its method and test it against a bar fixed in advance, on one cheap rig. The closest work is RoboSeed [26], which scores VLA and world-action models on five dimensions, including reproduction evidence and deployment affordability; its authors note that the score is not a probability that a result is correct. We give one verdict per claim on one rig, with the data behind it and the reason we think it failed.

10 Conclusion

We re-tested 36 claims from 21 robot-learning papers on a rig of about €1,500, with under $50 of cloud GPU. 12 held, and 10 failed with the authors' method on our data. The most useful part was the cheapest baseline: the same world model could pass one baseline and fail another, and 5 of the 11 methods we compared with one lost to a signal we had for free, although only 4 of these losses have an interval below zero, of which LeWorldModel's, DINO-WM's and ContactGuard's are clean cases with the authors' method. Our advice to anyone re-testing a paper on a cheap rig is to run that baseline first, on their own data, with the bar fixed before the test. Everything is public, and every number can be rebuilt from the repository. Authors who think a failure here is ours (Table 4) are welcome to tell us, and we will correct it.

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Appendix: every paper

paper	their rig	claim tested	bar	our result	verdict
LeWorldModel [9]	sim	A small JEPA world model learns the arm's dynamics well enough to plan with it.	Dynamics: beat copy-last, every corruption ≥ 1.3 at $K = 30$. Planning: rank the executed chunk first among 8 in ≥ 0.8 of anchors.	dynamics 0.31 vs 0.89 copy-last (reproduces); planning top-1 0.60 vs 0.68–0.90 for an action-only baseline (fail)	fail (authors' method)
DINO-WM [1]	sim	A predictor on frozen DINOv2 patches plans better than one on a learned latent.	G1b top-1 ≥ 0.8 at some target distance.	arm-pose R^2 0.92 (vs 0.57 for W1); G1b top-1 0.55–0.61; no-model baseline 0.68–0.90	fail (authors' method)
RoboJEPA [18]	>$20k	Pretrained RoboJEPA predicts my latents, uses the actions, and runs on the robot's computer.	9-step L1 ≥ 15 % below copy-last (300M) and ≥ 5 % below shuffled actions; 2-view encoder ≤ 200 ms on the Orin.	11.8 % below copy-last (300M; 1B 15.8 % on 30 clips); 9–10 % below shuffled; encoder 526 ms on the Orin NX	fail (authors' method)
H-JEPA [17]	sim	A slow second level makes latent distance track task progress.	Progress monotonicity Spearman $\rho \leq -0.80$; false stalls ≤ 10 %.	level-2 ρ -0.73; false stalls 52 %	fail (outside the paper's regime)
ContactGuard [15]	$5–20k	An event-anchored probe on imagined latents predicts grasp failure better than the current latent.	AUROC ≥ 0.85 and ≥ 0.05 above the current latent.	0.70 imagined vs 0.82 current	fail (authors' method)
FAIL-Detect and SAFE [2] [5]	>$20k	A conformal band on a learned failure score keeps false alarms at α while catching failures.	Successes flagged $\leq 1/20$; failures flagged ≥ 50 %.	9.1 % of successes flagged at α = 5 %; 24 % of failures caught (first band built wrong, see WS-214: 22.7 % / 46 %)	fail (our variant)
ReAP [19]	$5–20k	A risk head on an action-conditioned latent warns before failures.	$\geq 5/8$ failures caught at $\leq 1/20$ false interventions; action-conditioned iff true $\geq$ shuffled + 0.05.	v1 AUROC 0.93 came from time and length; v2 0.789 vs 0.708 shuffled, 3/8 at 5/42	fail (our variant)
VLA-Corrector [16]	sim	A verifier on observable history flags a failing rollout early enough to correct it.	$\geq 5/8$ failures at $\leq 1/20$ false interventions.	the paper's Learned Verifier on my rollouts, built and trained as the paper describes (54 RGB features per view, joint positions, executed actions, its labels, optimiser and gate): 1/8 caught at 3/42 false interventions, AUROC 0.53; the earlier run with the same input and our own training: 2/8 at 3/42, AUROC 0.59; cosine drift of the W1 latent 0/8, AUROC 0.30	fail (authors' method)
Robometer [7]	>$20k	A 4B reward model judges SO-101 rollouts zero-shot.	Pooled AUROC ≥ 0.85 and within-policy ≥ 0.8 on ≥ 5 of 7 policies.	pooled AUROC 0.818 on 2099 ArmnetBench rollouts; episode length alone 0.938	fail (authors' method)
SmolVLA [3]	<$5k	A 450M VLA succeeds on an SO-101 task and can run on low-cost hardware.	10-trial operator-judged gate (no numeric bar); chunk inference within the async budget (667 ms).	9/10 on bowls; 974 ms in PyTorch, 238 ms with vla.cpp on the Orin NX	reproduces
VLA-JEPA [6]	sim	VLA-JEPA trains cheaply on SO-101 data and runs on the robot.	10-trial operator-judged gate (no numeric bar).	9/10 for $9.7; 355 ms per chunk on the Orin NX	reproduces

Table 5: One row per paper with a claim of its own; FAIL-Detect and SAFE share a row. D-JEPA is left out: its only claim was run with proxy labels, not the paper's success labels, so it was never really tested. The bars are ours, written before each test, from the paper's own number where it gives one (that number is kept in the register). Internal names: W1 is our LeWorldModel and W2 our DINO-WM, both trained on our teleoperated data; names such as G1b, H0 or P1 are the ids of our tests in the repository. The rig price of €1,350 counts the parts with a stated price; 2 small parts are not priced yet. Read the verdict against the rig each paper was shown on.

paper	their rig	claim tested	bar	our result	verdict
REKI [21]	sim	Policies without memory oscillate between poses at a measurable rate.	$\geq$ 10 % of failed episodes oscillate (their criterion).	20.3 % of all episodes, 24.0 % of failures; REKI trained on our data: 19/20 + 19/20 on the arm, oscillates in 1/40	reproduces
Real-Time Chunking [4]	unstated	Conditioning a new chunk on the executing one removes the jump at chunk boundaries.	Best gradient-free method cuts the median jump $\geq$ 50 % against naive async, drift $\leq$ +10 %, within 667 ms.	prefix overwrite cuts the jump 71–73 % against naive async at +0.4 ms; tuned LeRobot identity mode is close	not proven
HWM (hierarchical latent world models) [10]	>$20k	A high-level world model over learned macro-actions beats the low-level model beyond about 1 s (their Fig. 6 crossover).	Crossover in [1.0, 2.0] s; at 2 s high-level error $\leq$ 0.95 $\times$ low-level; learned macro-actions $\geq$ net joint displacement on cosine.	no crossover: W1 0.382 $\pm$ 0.005 $\rightarrow$ 0.563 $\pm$ 0.007, high level 0.769 $\pm$ 0.009 $\rightarrow$ 0.759 $\pm$ 0.009 (0.73–2 s); cosine 0.608 $\pm$ 0.007 vs 0.551 $\pm$ 0.006	not proven
LeJEPA identifiability [12]	sim	A SIGReg JEPA trained on goal-directed data recovers the true latent state only partially and unevenly.	W1's mean R^2 from latent to joints $<$ 0.80; anisotropy (max $-$ min per-joint R^2) $\geq$ 0.20; the higher-R^2 model has the higher G1b top-1.	W1 mean R^2 0.565, anisotropy 0.639 (shoulder lift 0.91, wrist flex 0.27); frozen DINOv2-S 0.867 / 0.301	reproduces
SURVEYOR [25]	sim	A threshold on the imagined-vs-reached latent distance accepts good plans and rejects bad ones ($\tau = 0.20$).	τ below the 10th percentile of non-equivalent pairs and $\leq$ 0.40; as a monitor $\geq$ 5/8 failures at $\leq$ 2/42; $\geq$ 0.05 AUROC over a shuffled drafter.	τ = 1.249 on W1 latents of 69 real rollouts; as a monitor 3/8 at 6/42 (S = 30), AUROC 0.756 vs shuffled 0.560	fail (authors' method)
Adaptive Action Chunking (AAC) [11]	>$20k	Chunk length chosen from the entropy of N sampled chunks follows the task's phases (long in transport, short at the grasp), lowers the boundary jump, and costs little with batched sampling.	median h* in transport $\geq$ grasp window + 2; jump with AAC + overwrite $\leq$ fixed + overwrite at d = 8; an N = 20 decision $\leq$ 298 ms on the Orin.	h* 2 in transport and 2 at the grasp (86 % of decisions at the floor); jump 0.973 vs 0.940; N = 20 decision 4,751 ms (sequential) on the Orin	fail (authors' method); the latency claim: our engine (no batched sampling)
PACT-VLA [22]	$5–20k	Picking among N sampled chunks by their flow residual under the pre-commit observation fits what happens next better than a random or stale pick, with no added control-step delay.	PACT beats a random pick and the stale-observation residual at d = 8 (CIs $<$ 0); its delay on the Orin $\leq$ the naive delay; at that delay it still beats a single sample.	offline $-0.27°$ vs random and vs the stale residual (both CIs $<$ 0); on the Orin 1,200 ms per cycle, 37 control ticks vs 8	fail (our engine)
Horizon $\times$ depth (Euler steps vs execution horizon) [23]	>$20k	With long open-loop commitment, cutting a flow-matching VLA's Euler steps from 10 to 2 contracts the action cloud around an unchanged centroid, saves most of the inference time and avoids edge-GPU thermal throttling.	median R(2)/R(10) $\leq$ 0.70 with the centroid within 0.05 relative and 3°; N = 2 latency $\leq$ 0.85 $\times$ N = 10 on the Orin, N = 10 within 5 % of 238 ms.	R(2)/R(10) 0.78 (CI 0.73–0.81), centroid 0.018 and 2.35°; 196.1 vs 237.5 ms per chunk (0.83, vision 151 ms); no throttling at N = 10 on the Orin, at MAXN_SUPER or 25 W	not proven
REMORA [24]	>$20k	Reading a progress distribution out at the supported peak nearest the previous estimate steadies the estimate and lets it fall during real setbacks, with no worse detection of the stages.	Against per-frame argmax: premature not higher and missed $\leq$ +2 points (P1); flips halved, CI upper $\leq$ 0.7 (P2); a decrease inside $\geq$ 50 % of the reversal windows with $\leq$ 20 % false regressions (P3).	premature 0.22 vs 0.25, missed equal (P1 pass); flips 0.70 $\times$ argmax, CI 0.48–0.94 (P2 fail); setback recall 2/7, false 0.52 (P3 fail); MAE 0.184 vs median 0.218	fail (our variant)

Table 5, continued.